A growing epithelial spheroid in an extracellular matrix: the Plexus2 implementation

`prototype/ecm` · runs in `log/okuda_ECM` · 6 August 2026

1. Sets, operators, schedule

A Plexus model is sets × fields × operators × schedule. The biological hierarchy here is four entities and one field; the basement membrane is a *specialised* ECM, not the ECM:

entity	set	representation	principal operators
epithelium	`vertex`, `cell`	3D active vertex model	`shape_energy_3d`, `morphogen_growth_3d`, `di`
junction	relation on the mesh	half-edges, keyed state	`junction_myosin`, `reconnect_t1_3d`
basement membrane	`basement_membrane_particle`	MPM + explicit bonds	`.._seed/bond/remodel/bond_break`, `integr`
interstitial ECM	`mpm_particle`	MPM fibres	`ecm_seed`, `ecm_stress`, `cell_to_ecm`
(field)	`mpm_grid`	MLS-MPM background grid	`mpm_scatter/gather/grid_update`

Every particle body scatters into and gathers from the *same* grid; that sharing *is* the mechanical coupling between them, so no pairwise contact model is written. Operators carry one of seven kinds. The four dynamics kinds return a per-set delta the engine integrates once per tick; `field`, `rewire` and `structural` mutate the grid, the relation E, or the membership $|S|$ in place and *return nothing.* That last fact drives a design decision in §5.

2. The spheroid: 3D active vertex model

Cells are faces of a closed triangulated shell; cell f's volume is the wedge from the tissue centre, $v_f = \frac{1}{3}\,\mathbf{c}_f\cdot\mathbf{N}_f$. The energy is

$$E \;=\; \sum_f \Big[K_A\,(A_f - A_f^0)^2 + K_P\,(P_f - P_f^0)^2 + \tfrac{1}{2}\Gamma P_f^2 + K_V\,(v_f - v_f^0)^2\Big] \;+\; \Lambda \sum_e m_e\,\ell_e \;+\; K_R \sum_i \big(|\mathbf{x}_i| - R_0\big)^2, \quad (1)$$

relaxed overdamped each frame, $\mathbf{x} \leftarrow \mathbf{x} - \eta\mu\,\nabla E$ with a per-step cap. The factor m_e is new (§5); with $m_e \equiv 1$ Eq. (1) is the stock Okuda energy and every prior run is bit-identical. Growth scales each cell's targets by a cumulative s_f,

$$s_f \leftarrow s_f\big(1 + \lambda(\rho + \mathrm{Hill}(a_f))\big), \qquad A_f^0 = A_f^{0,\mathrm{init}} s_f^2, \quad P_f^0 = P_f^{0,\mathrm{init}} s_f, \quad v_f^0 = v_f^{0,\mathrm{init}} s_f^3, \quad (2)$$

and `divide_3d` splits a cell once v_f passes twice its reference. From 200 cells the tissue reaches ~ 6000 and its apical radius $4.66 \to 16.6$ (arbitrary units) in 402 frames.

3. The ECM: MLS-MPM

The matrix is material points on a background grid, fixed-corotated with $F = RS$:

$$\boldsymbol{\tau} \;=\; 2\mu\,(F - R)F^{\top} + \lambda J(J-1)\,I, \qquad \boldsymbol{\sigma} \;=\; \boldsymbol{\tau}/J, \qquad \sigma_{\mathrm{vM}} = \sqrt{\tfrac{3}{2}\,\mathbf{s}{:}\mathbf{s}}, \;\; \mathbf{s} = \boldsymbol{\sigma} - \tfrac{1}{3}\mathrm{tr}(\boldsymbol{\sigma})I. \quad (3)$$

$\boldsymbol{\tau}$ was already computed every substep to build the affine momentum matrix and then discarded; `mpm_scatter` now optionally caches the Cauchy stress $\boldsymbol{\sigma}$ to a per-particle buffer (`store_stress`, off by default, byte-identical when off). This matters quantitatively: colouring by the volumetric measure $|J-1|$ put 0.6% of the matrix above the display floor, whereas σ_{vM} of the *same* run puts 64% there. A matrix pushed by a growing sphere *shears*; the fixed-corotated law resists volume change far more stiffly than shape change, so the volumetric measure was blind to the deformation, not the mechanics.

4. Spheroid ↔ ECM

The two solvers cannot share a world. `mpm_grid` is hard-coded to $[0,1]^3$; the epithelium lives in a 50-unit box, and rescaling it there was tried and measured, not assumed — the same run gives $200 \to 319$ cells at its own scale but $200 \to 212$ rescaled, because the terms of the vertex energy carry different powers of length and a $50\times$ shrink changes which one wins. So the two run as two passes joined by one geometric scale, applied to a *recorded* shape.

That recording is the whole coupling, and it runs one way. The epithelium's surface is a moving boundary the matrix feels; nothing the matrix does reaches back. Fibres inside the surface are pushed out by a soft penalty and then, as a backstop, projected out:

$$\mathbf{a}_i \;=\; k_c\,\big(R(\hat{\mathbf{u}}_i, t) - r_i\big)_+\,\hat{\mathbf{u}}_i, \quad (4)$$

and the reaction is binned by direction and divided by the area each bin covers, $R^2\,d\Omega$, to give a pressure map $P(\theta,\phi,t)$. A cell facing a stressed matrix then grows more slowly:

$$g \;=\; f + \frac{1-f}{1 + \big(P/p_{1/2}\big)^n}, \qquad s_{\mathrm{now}} \leftarrow s_{\mathrm{prev}}\,\big(1 + g\,(f_{\mathrm{grow}} - 1)\big). \quad (5)$$

Because it is the growth *rate* that is gated, a small difference compounds over 400 frames, and a 2.75× stress anisotropy becomes a shape: an oblate cavity takes the tissue to oblateness 1.43 where a flat pattern of the same strength leaves it round at 1.015.

5. The two new levels

Junctions. In the vertex model contractility is one number for the whole tissue. Each junction now carries its own myosin m_e, multiplying the line tension, and chasing a setpoint on a timescale τ_m:

$$\Lambda \sum_e \ell_e \;\longrightarrow\; \Lambda \sum_e m_e\,\ell_e, \qquad m_e^{ss} = a\Big(1 + \beta\left(x_e/\langle x\rangle - 1\right)\Big), \qquad \dot{m}_e = \left(m_e^{ss} - m_e\right)/\tau_m. \tag{6}$$

What x_e should be is the substance. Our first choice was junction *length*, which makes the loop *longer* → *more myosin* → *shorter* — a negative feedback that evens junction lengths out. The measured biology is the opposite: myosin is recruited by *tension* and is enriched on *disassembling* junctions, giving a positive feedback that drives neighbour exchange. Worse, in the fast-myosin limit — the correct one, since myosin turns over in seconds — substituting the setpoint turns Eq. (6) into a line tension plus a harmonic spring on each edge, so "β reduces the spread of junction lengths" is that spring's definition and cannot be evidence for it. x_e is now the actual edge tension, $\Lambda m_e + 2K_P(P_f - P_f^0) + \Gamma P_f$, taken from the same energy the mechanics minimises; and the observable we report is the **T1 rate**, which the two feedbacks move in opposite directions where the length spread does not.

Basement membrane. A sheet of nodes joined by explicit crosslinks — MPM particles alone have no connectivity, so they cannot be crosslinked and cannot fragment. Each crosslink is a spring that forgets being stretched on a timescale τ_r and breaks past a strain ε_b; each node is tied to a fixed *direction* on the recorded surface, which is what makes it adhesion rather than contact, since anchoring to the nearest point would let the node slide and the sheet would never stretch:

$$\mathbf{f}_{ij} = k_b\,(L_{ij} - \ell_{ij}^0)\,\hat{\mathbf{d}}_{ij}, \qquad \dot{\ell}_{ij}^0 = (L_{ij} - \ell_{ij}^0)/\tau_r, \qquad \text{bond dies if } (L - \ell^0)/\ell^0 > \varepsilon_b, \tag{7}$$

$$\mathbf{a}_i = \kappa\big(\mathbf{c} + \hat{\mathbf{u}}_i(R(\hat{\mathbf{u}}_i, t) + \delta) - \mathbf{x}_i\big), \qquad \hat{\mathbf{u}}_i \text{ frozen at seeding.} \tag{8}$$

Remodelling is not optional: a purely elastic sheet cannot enclose a surface that grows 260% in linear strain, and it tore at every ε_b we tried. With it, the sheet settles at $\langle \epsilon \rangle \simeq \dot{\varepsilon}\,\tau_r$. And because area grows as R^2 while deposition adds a fixed fraction of existing material per frame, secretion has a threshold, $\rho^* = 2\ln(R_1/R_0)/N$ — though that follows from the operator's own setpoint, so measuring it confirms an implementation rather than testing a hypothesis.

Two corrections. The sheet was integrated with a *mass*, and everything this section used to report about oscillation, critical damping, tracking lag and a stiffness ceiling followed from that; at $\mathrm{Re} \sim 10^{-10}$ the motion is $\gamma\dot{\mathbf{x}} = \mathbf{F}$ and none of it exists. It is now overdamped. And the metric these claims rested on — the largest connected component — was `NaN` on every frame any analysis read, so it had never once been reported; it is now carried forward beside the mean degree z, which must exceed ≈4 for the sheet to be mechanically a sheet at all.

Finally, on naming: `integrin_adhesion` is a tether to a surface, not a receptor. In mammals $\alpha_1\beta_1$ and $\alpha_2\beta_1$ do bind collagen IV, so the receptor exists — but the dominant epithelial anchor is to *laminin*, which is not modelled, and the sheet has no attachment to the stroma either. Removing the tether therefore leaves it attached to nothing, which is why that run is a control and not a knockout.

6. The two schematics

The two levels added on top of the spheroid and the matrix. Everything else about them — the configurations, the measurements, and the external audit that found much of it wanting — has moved to `spheroid_ecm_archive.tex` while the model is being corrected.

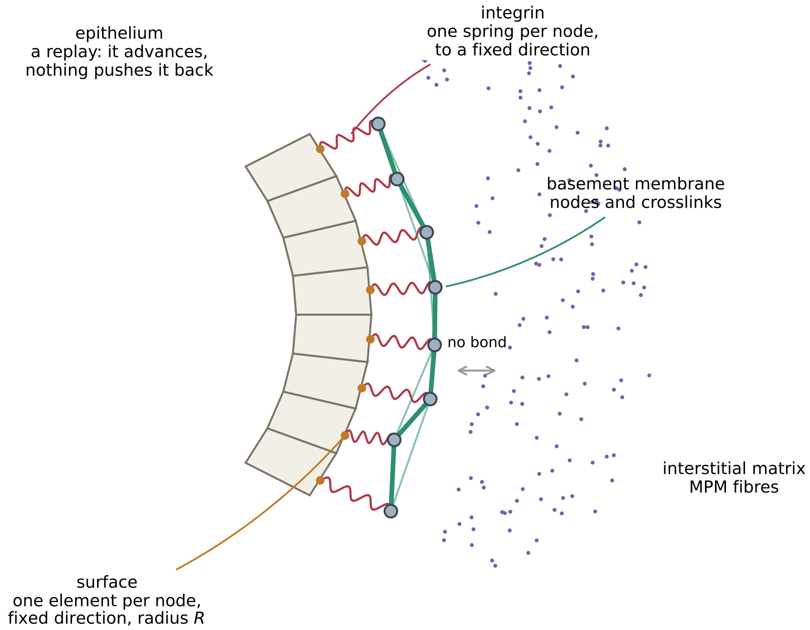

Figure 1: **What the membrane is made of.** Four objects, and the arrows between them are the whole model. The *epithelium* (cream wedges) is a recording: it advances on a prescribed schedule and nothing can push it back. Its outer boundary is the `surface` Level (orange), one element per membrane node, each holding a fixed direction $\hat{\mathbf{u}}_i$ and the radius $R(\hat{\mathbf{u}}_i, t)$ the tissue reached in that direction. The *basement membrane* is grey *nodes* joined by green *crosslinks*, sitting a distance δ outside. Each node is tied to its *own* surface element by one red *integrin* spring—one per node, to a fixed angular position, which is what makes it adhesion rather than contact. Outside is the interstitial matrix (purple), which the membrane never bonds to: they meet only through the shared grid, or in the spring-graph version, not at all.

Figure 2: **What a junction is, and the one loop that makes it interesting.** Left: two cells share an edge, and that edge *is* the junction—it belongs to neither cell, which is why its state is keyed by the unordered pair of vertices (i, j) rather than stored on a cell. Myosin m_e sits on the shared edge and sets its tension. Right: the feedback. A junction longer than average recruits more myosin (β), which pulls harder, which shortens it. That loop is the only thing here that Okuda's model cannot already do.